



兰州大学
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Data Modeling and Visualization Report

Course name: INFO442 Data Science Projects

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Date : 2026. 6. 27

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1 Project Overview and Objectives

This project aims to build a customer repurchase prediction model for an online retailer. The core objective is to identify customers who are most likely to make another purchase within the next 30 days, using historical transaction data. This enables data-driven decisions for targeted marketing, customer relationship management, and inventory planning.

1.1 Project Goals

1. **Prediction Target:** Predict whether a customer will make a purchase within the next 30 days (binary classification problem).
2. **Business Value:** By identifying high-repurchase-probability customers, the marketing team can design more targeted promotional strategies, improve marketing ROI, and better understand factors influencing customer retention.
3. **Data Source:** The `online_retail_cleaned.csv` dataset, containing real online retail transaction records from December 2010 to December 2011.

2 Methodology

2.1 Data Preprocessing and Feature Engineering

Prior to model building, we performed necessary data cleaning and feature engineering to construct a predictive feature set. Key steps include:

1. Data Cleaning:

- Removed return transactions (`IsReturn == False`).
- Ensured `Quantity` and `UnitPrice` are positive values.

2. Customer Segmentation Features:

- Product Clustering:** Used TF-IDF vectorization on product descriptions and applied K-means to create 13 product categories (Cluster). Cluster labels reflect customer purchasing preferences.
- RFM Features:** Calculated classic RFM metrics based on customer transaction history:
 - Recency: Days since the customer's last purchase relative to a reference date.

- **Frequency:** Total number of orders placed by the customer during the training period.
- **Monetary:** Total amount spent by the customer during the training period.
- **Statistical Features:** Computed features such as average order value, max/min order value, number of unique products purchased, total quantity purchased, average quantity per transaction, and standard deviation of quantity to provide a more comprehensive description of purchasing behavior.
- **Derived Features:** Created composite features such as Monetary_per_Freq (average spend per order) and Qty_per_Trans (average quantity per transaction).

3. Label (Target Variable) Generation:

Defined the "repurchase" label as: whether the customer made at least one new purchase within 30 days after the end of the training period. This forms a standard binary classification problem.

2.2 Model Selection and Justification

This report selects and compares two classic models widely applied in customer prediction problems:

1. Logistic Regression:

- **Rationale:** Logistic Regression is a linear model with excellent interpretability. By examining model coefficients, we can clearly understand the direction and relative importance of each feature's impact on repurchase probability. This is crucial for explaining predictions to business stakeholders and formulating marketing strategies.
- **Implementation:** Used `sklearn.linear_model.LogisticRegression` with `class_weight='balanced'` to address potential class imbalance in the data.

2. Random Forest:

- **Rationale:** As an ensemble learning method, Random Forest can capture non-linear relationships and complex feature interactions. It generally offers stronger predictive power than Logistic Regression and is insensitive to feature scaling. Additionally, it provides feature importance scores to identify the most significant predictors.
- **Implementation:** Used `sklearn.ensemble.RandomForestClassifier`, also with `class_weight='balanced'`, and controlled model complexity using parameters such as `n_estimators=200` and `max_depth=10` to prevent overfitting.

2.3 Model Evaluation Approach

Due to the dataset containing only one year of data, rigorous time-series validation (i.e., training on earlier data to predict later data) was not feasible. To demonstrate the evaluation process, we applied a random split method, dividing the processed customer data into training (80%) and validation (20%) sets.

Limitation: Random split ignores the temporal order of data, potentially causing data leakage (using future information to predict the past), which may overestimate model performance in real business scenarios. This limitation is discussed further in the limitations section.

3 Analysis and Results

3.1 Model Performance Comparison

The following table summarises the performance of Logistic Regression and Random Forest models on the validation set.

Model	ROC-AUC	F1-Score	Accuracy
Logistic Regression	0.7535	0.5874	0.7252
Random Forest	0.7409	0.5784	0.7240

- **ROC-AUC:** Logistic Regression (0.7535) slightly outperforms Random Forest (0.7409), indicating a marginally better overall ability to distinguish between repurchasers and non-repurchasers.
- **F1-Score:** Logistic Regression (0.5874) also surpasses Random Forest (0.5784). This suggests better balance between Precision and Recall. Given the business objective of identifying potential repurchasers (positive class) from a large customer base, F1-Score is a critical metric as it addresses both prediction accuracy and coverage.
- **Accuracy:** Both models achieve similar accuracy (~72.5%).

Conclusion: Under this dataset and experimental setup, Logistic Regression demonstrates slightly superior overall performance. Its higher AUC and F1-Score indicate it is a more robust choice.

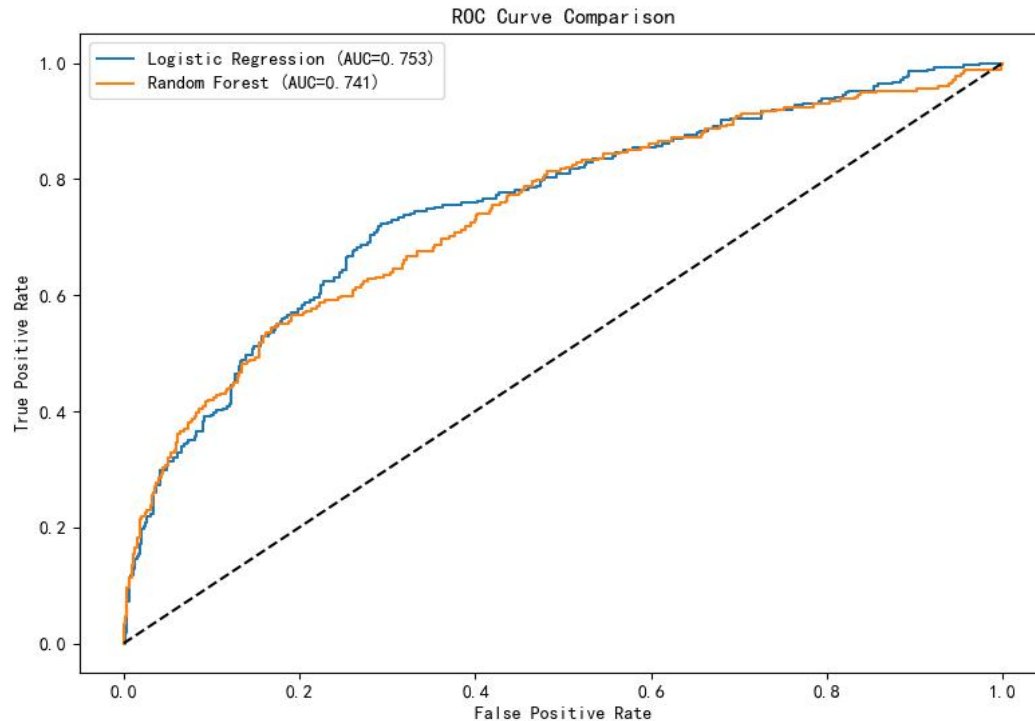
3.2 Model Evaluation Visualisation

ROC Curve Comparison

The ROC curve (Receiver Operating Characteristic Curve) visually illustrates model performance across different

thresholds. A larger Area Under the Curve (AUC) indicates better model performance.

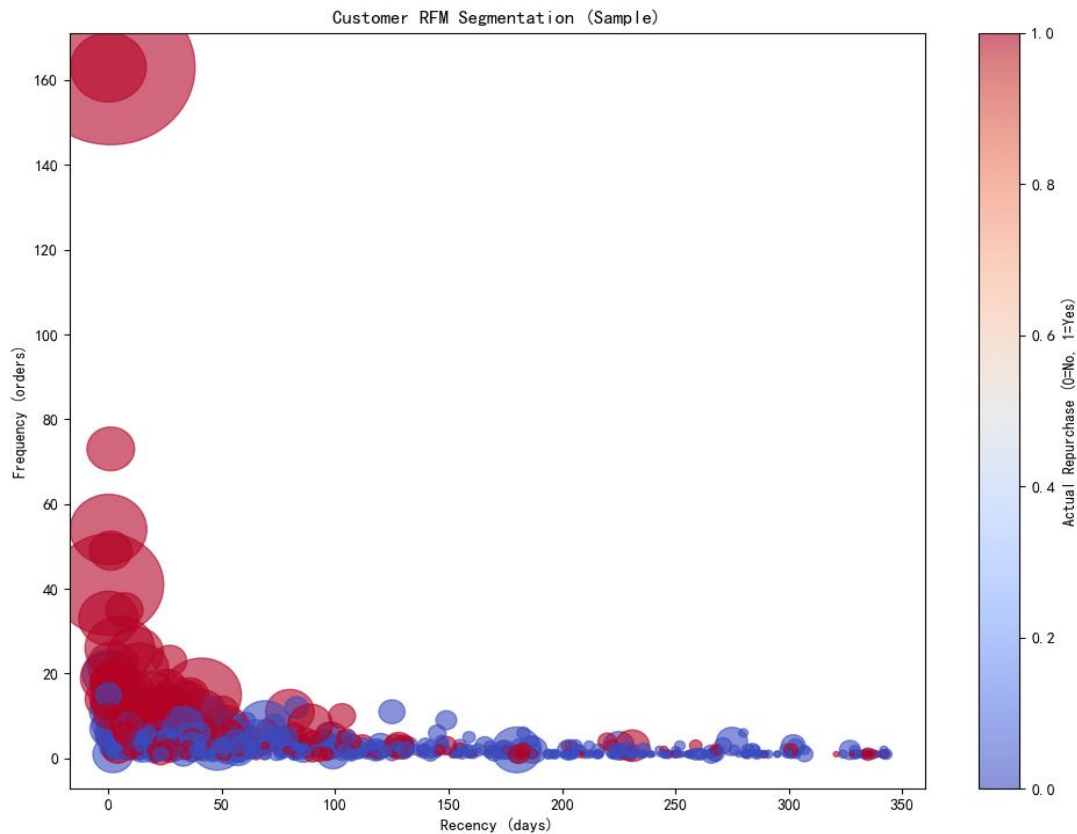
The figure below shows the Logistic Regression ROC curve (blue) positioned slightly above the Random Forest curve (orange), further validating the marginal advantage of Logistic Regression in discriminative ability.



3.3 Customer Value Pyramid: Predicted vs Actual

This visualisation segments customers into four tiers (Value Tier) based on their Monetary value (from low to high) and compares the actual repurchase rate with the predicted repurchase probability from the Random Forest model for each tier.

- **Observation:** As customer value increases (from "Low" to "High"), both actual repurchase rates and predicted repurchase probabilities show a clear upward trend. High-value customers exhibit significantly higher repurchase rates than low-value customers.
- **Insight:** This aligns with classic RFM model theory: "the more a customer spends, the more likely they are to repurchase." The model accurately captures this business pattern.
- **Application:** Marketing resources should be directed toward "Medium-High" and "High" value tier customers, as they not only contribute higher revenue but also demonstrate stronger retention tendencies.



3.4 Feature Importance Analysis

This figure displays the top 10 most important features from the Random Forest model.

• **Key Features:** Monetary (total spend), Frequency (purchase frequency), Monetary_per_Freq (average order value), and Recency rank at the top.

• **Insights:**

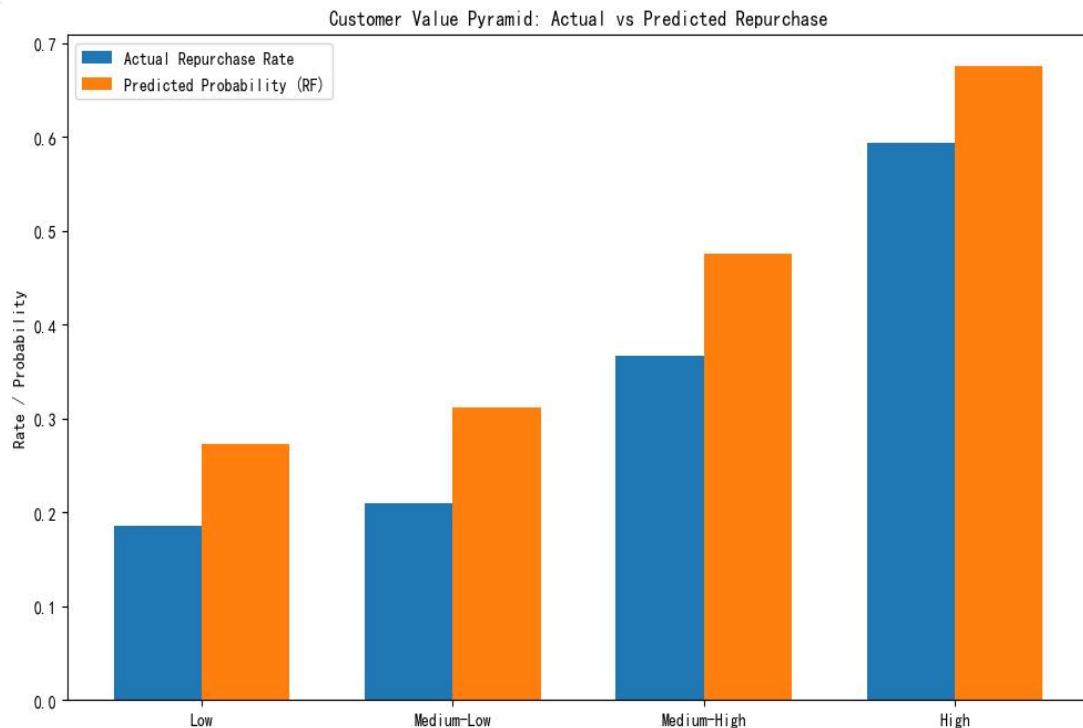
Monetary & Frequency: Historical spend and purchase frequency are the strongest predictors. High-spending, high-frequency customers are more likely to repurchase.

Recency: The more recent the last purchase (lower Recency value), the higher the repurchase likelihood, validating the marketing principle that "recent behaviour predicts future behaviour."

Avg_OrderValue & Max_OrderValue: Order value features are also important, indicating that spending capacity itself is a key differentiator.

Num_Products: Customers purchasing a wider variety of products tend to have higher repurchase propensity.

Cluster: Product cluster features also enter the top ten, suggesting purchase preferences have some predictive power for repurchase, though their relative importance is lower.



4 Model Limitations and Potential Failure Modes

This analysis has several limitations that could lead to model failure in future real-world applications:

1. Lack of Temporal Validation:

- **Issue:** As previously noted, with only one year of data and a single calendar year, we could not perform rigorous time-series cross-validation, instead using random split.
- **Potential Failure:** Time-series data contains trends and seasonality. Random split may allow the model to use "future" information to predict the "past," overestimating generalisation performance. When applied to truly future data, model performance may degrade significantly.
- **Mitigation:** Use TimeSeriesSplit or hold out the last consecutive period (e.g., final 1-2 months) as a validation set to more realistically simulate the business scenario.

2. Limited Data Representativeness:

- **Issue:** The data covers only one year, potentially failing to capture long-term customer behaviour patterns or year-over-year seasonality (e.g., Christmas shopping season).
- **Potential Failure:** The model may overfit this specific year's consumption patterns. For instance, if the 2011 Christmas promotional campaign was particularly aggressive, the model would overestimate repurchase rates each December, failing in regular years.

- **Mitigation:** Collect multi-year data to train more robust models.

3. Class Imbalance Handling:

- **Issue:** Although we used `class_weight='balanced'`, repurchasers are typically the minority class. If class imbalance is extreme, the model may bias toward predicting "non-repurchase."
- **Potential Failure:** The model may miss many potential high-value repurchasers, leading to lost marketing opportunities.
- **Mitigation:** Explore more advanced sampling techniques (e.g., SMOTE) or adjust the decision threshold (Threshold Tuning) to achieve a more business-aligned balance between Precision and Recall..

5 Conclusion and Next Steps

Conclusion:

This project successfully built and evaluated two models for predicting customer repurchase. Analysis shows that Logistic Regression slightly outperforms Random Forest on this dataset, achieving an AUC of 0.75. We found that customers' historical spend, purchase frequency, and recency are the most critical factors for predicting future repurchase. Through the Customer Value Pyramid analysis, we further confirmed that high-value customers exhibit higher retention tendencies, providing a foundation for targeted marketing.

Next Steps:

- 1. Improve Validation Strategy:** Adopt rigorous time-series split methods to evaluate models and obtain more reliable performance estimates.
- 2. Enhance Feature Engineering:** Explore additional time-series features such as standard deviation of purchase intervals, days since last purchase relative to training period end, et
- 3. Deploy and Monitor:** Deploy the best-performing model into a business system, continuously monitor its performance, and periodically retrain with new data.
- 4. A/B Testing:** Based on model predictions, design targeted marketing A/B tests across different customer segments to quantify the model's actual business value.